

Rectangle incidence circuits for a stronger integer-multiplication exponent

Douglas Colkitt*

October 7, 2026

Abstract

We replace individual neighbor channels in the finite bit network of OpenAI’s integer-multiplication manuscript by rectangle circuits with shared intermediate computations. Non-degenerate common frames preserve the absolute rank deficit. Together with nonadjacent axis routing and the previously audited parameter recipe, the construction supports the conditional saving $\kappa = 2^{-67}$, a factor 256 above our preceding 2^{-75} witness. The complex network is unchanged. No practical runtime improvement or global optimality is claimed.

1 Scope and construction

The model is a fixed finite-alphabet Turing machine with a fixed finite number of one-dimensional tapes. We assume the unchanged algorithmic interfaces of *Integer multiplication below $n \log n$* , pinned at

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

The construction below replaces the bit primitive. The remaining parts of the manuscript’s multiplication theorem are assumed.

1.1 A rectangle circuit for the bit network

Retain $h = 46$, $\mathcal{T} = \binom{[h]}{3}$, $v = 15180$, $N = v^3$, and $m = h^3 = 97336$. We change only the bit network. The complex network and its guard constants are unchanged. On $F = \mathbb{Q}^h$ use the form $I - J/9$; the indicator t_T has norm two, and neighboring indicators are orthogonal.

Every ordered neighbor pair has a unique common point i . Removing i leaves an ordered pair of disjoint two-subsets of the other 45 points. Partition this disjointness relation into rectangles $\mathcal{S} \times \mathcal{U}$. In each rectangle every source subset is disjoint from every target subset. Restoring i gives source and target triple families, all cross pairs of which are neighbors.

Here is an explicit finite procedure for the partition used below. Let $D_{a,b}(n)$ be the relation between disjoint a - and b -subsets, with $0 \leq a, b \leq 2$. For a partition let (C, L, R) count its rectangles, source memberships, and target memberships. An empty relation contributes $(0, 0, 0)$; if $a = 0$ or $b = 0$, one full rectangle contributes $(1, \binom{n}{a}, \binom{n}{b})$. Available nontrivial choices are singleton-row stars, singleton-column stars, or any split $n = l + (n - l)$, $1 \leq l < n$. For a split, separate pairs by their source and target sizes in the first part. Each case is a product $D_{x,y}(l) \times D_{a-x,b-y}(n - l)$.

*Prepared with assistance from OpenAI Codex. This is not independent review or endorsement by OpenAI.

Product partitions multiply the three counts coordinatewise; disjoint cases add them. Thus the procedure always produces an exact partition.

Select a plan by minimizing $\alpha L + \beta R - C$, initially $\alpha = \beta = 1$. A product with a complete factor having counts $(1, L_0, R_0)$ passes weights $(\alpha L_0, \beta R_0)$ to the other factor. If both factors are nontrivial, both are $D_{1,1}$: try for each its selected unweighted plan, row stars, and column stars, and take the least product cost. Try all split sizes and the two star alternatives; ties retain the first plan in the order just specified, with row stars before column stars and increasing split sizes. This is a finite recursive construction, not a global optimality claim. The companion integer implementation and exhaustive pair check give

$$(C, L, R) = (2759, 27278, 27534), \quad L + R - C = 52053.$$

Its rectangles contain each of the $\binom{45}{2}\binom{43}{2} = 893970$ ordered disjoint pairs exactly once.

For a rectangle with a sources and b targets allocate $a + b - 1$ auxiliary roles: a input slots and b output slots sharing one pivot. Let L now denote the invertible bit mixer that adds the other input slots into the pivot and then adds the pivot into the other output slots. Its output-by-input submatrix is all ones. Its inverse reverses these two operations. Let V copy sources into their input slots and J inject output slots into their targets. Then JLV is exactly this rectangle's neighbor contribution. Use separate roles for distinct rectangles. The side-role count per invocation is

$$R_{\text{side}} = 46(52053) = 2394438.$$

1.2 Restoration and common frames

Let G, R denote the original central gather and scatter. The forward chronological schedule is

$$L, J, L^{-1}, R, V, G, R, L, J, L^{-1}, G, V.$$

Here L is applied independently on each rectangle and all other operations are their indicated bit additions. For arbitrary initial side vector z , the first three operations add JLz to the target and restore z . The copy changes it to $z + Vx$; the later mixer and injection add $JL(z + Vx)$. Their dirty-scratch contributions cancel, and the last inverse mixer and copy restore z . The centers contribute RGx and are also restored. Since JLV is precisely the neighbor matrix, $(JLV + RG)x = x$. Thus the invocation implements $y \leftarrow y + x$ on arbitrary inputs. Reversing and inverting its gates implements the inverse invocation.

At stage j write, as in the original tensor construction,

$$A = F^{\otimes(j-1)} = B \perp P, \quad P = \langle t_{a_1} \otimes \cdots \otimes t_{a_{j-1}} \rangle, \quad Q = \langle t_{a_{j+1}} \otimes \cdots \otimes t_{a_3} \rangle.$$

Suppress the future factor Q and abbreviate the following labels:

$$\begin{aligned} D_0 &= B \otimes F, & D_1 &= A \otimes F, \\ X_t &= (B \otimes F) \perp (P \otimes \langle t \rangle), & Y_t &= (B \otimes F) \perp (P \otimes t^\perp), \\ D_U &= (B \otimes F) \perp (P \otimes U). \end{aligned}$$

For a forward rectangle U is the span of its physical X indicators. Every indicator contains its common point i . For any vector in this span, $\sum_j u_j = 3u_i$, whence

$$\langle u, u \rangle = \sum_{j \neq i} u_j^2 > 0 \quad (u \neq 0).$$

Indeed zero outside coordinates also force $u_i = 0$. Thus U is nondegenerate, including when the ambient form is indefinite. Orthogonality of all cross pairs gives $\langle t_X \rangle \subset U \subset t_Y^\perp$.

The common frames of the forward gates, in the displayed chronological order, are

$$D_0, D_0, D_0, D_0, X_{t_X}, D_1, D_0, D_U, Y_{t_Y}, D_1, D_1, D_1.$$

The gates at X_{t_X} and Y_{t_Y} are grouped separately by data triple; the middle mixers at D_U are grouped by rectangle. The early side injection is instead at the common label D_0 , allowing the earlier scratch mixers. This is a monotone enlargement of the incoming physical Y label.

For stage 2 reverse and invert the scalar schedule with logical source Y and target X . Its operations and common frames are respectively

$$\begin{aligned} & V, G, L, J, L^{-1}, R, G, V, R, L, J, L^{-1}, \\ & D_0, D_0, D_0, X_{t_X}, D_U, D_1, D_0, Y_{t_Y}, D_1, D_1, D_1, D_1. \end{aligned}$$

Now U is the span of the physical X indicators, which are the rectangle's logical targets. The same nondegeneracy and orthogonality proof applies.

All frames above are tensored with Q . An input-only side role follows $D_0 \subset X_t \subset D_U \subset D_1$ in a forward invocation; an output-only role follows $D_0 \subset D_U \subset Y_t \subset D_1$; the pivot follows both. In the reverse invocation the two roles exchange these physical descriptions. All side edges are therefore increasing, with nondegenerate orthogonal residuals. Physical X still changes from $A \otimes \langle t \rangle \otimes Q$ to $(A \otimes F) \otimes Q$, and physical Y from $B \otimes \langle t \rangle \otimes Q$ to $Y_t \otimes Q$. These are the original interstage boundaries. Each central role has one decreasing edge, from D_1 to D_0 , losing dimension h . Thus the total decreasing dimension remains

$$L_{\text{dec}} = 3v^2h^2.$$

1.3 Sharing all first-stage and third-stage auxiliary roles

Pair the ground points consecutively. Define a permutation π of triples as follows. A triple containing a full pair and a singleton keeps the singleton and cycles its full pair among the pairs other than the singleton's pair. A triple using three different pairs keeps the point in the least indexed pair and flips the other two points to their partners. The first rule is a cycle on each fixed-singleton class and the second an involution on the remaining classes. Hence π is bijective and $|T \cap \pi(T)| = 1$.

Match the whole auxiliary bank of the stage-1 invocation with fixed triples (A, B) to that of stage 3 with fixed triples $(B, \pi(A))$, keeping the local role index. Both invocations restore arbitrary auxiliary inputs, and the matching is bijective. No simultaneous roles are identified. It includes centers as well as side roles.

Every auxiliary first gate now has label $D_0 \otimes Q$ and every last gate has label $D_1 \otimes Q$. In a matched pair the joining labels are

$$E = F \otimes \langle t_A \rangle \otimes \langle t_B \rangle, \quad H = \langle t_B \otimes t_{\pi(A)} \rangle_{F \otimes F}^\perp \otimes F.$$

Both are nondegenerate and $E \subset H$, because $t_A \perp t_{\pi(A)}$. The edge $E \rightarrow H$ replaces $E \rightarrow F^{\otimes 3}$ and $0 \rightarrow H$, reducing total projection-difference rank by exactly m per removed role. No additional decreasing edge arises. The final role count and rank sum are

$$\begin{aligned} W_b^{\text{rect}} &= 2N + 2v^2(R_{\text{side}} + h) = 1110529317427200, \\ s_b^{\text{rect}} &= W_b^{\text{rect}}m - N + 2L_{\text{dec}} = 108094481068699857600, \\ W_b^{\text{rect}}m - s_b^{\text{rect}} &= 572394081600 > 0. \end{aligned}$$

The other margins are $g_1 > 1/2$, $g_4 = 20a/21$, $g_5 = 13/112$, $g_6 = 299/336$, and $g_7 = 1/21$, all strictly larger. The positive gap absorbs the fixed logarithmic factors as in the existing strict-margin argument, yielding $O(n(\log n)^{1-2^{-67}})$ conditionally.

The dimension exponent, stopping threshold, complex network, and guard analysis are retained: the guard is $p^{20/21} = o(p)$, the new chunk has $K = \Theta(p^{3a/70}) = o(\ell)$ with $\ell = \Theta(p^{20/21})$, and $K/\log p \rightarrow \infty$. The numerical maps and final rounding are unchanged. No use is made of the separate recurrence refinement requiring $\sigma \leq \tau$; the original separate layer inequalities pass here.

3 Reproduction and boundary of verification

The companion artifacts are:

- `scripts/incidence_rectangles.py` and `scripts/incidence_network.py`;
- `certificates/incidence-network.json`;
- `patches/h46-incidence-67.patch`.

The rectangle checker enumerates every ordered disjoint pair at the target size. Small forward and inverse invocations are checked on every input basis vector, including every scratch input. A full three-stage small network checks bank exchange and restoration with shared nonzero scratch. Rational tests verify the norm identity and sharing inclusion. The written frame argument and the unchanged upstream machinery remain mathematical proof obligations; these checks are not a formal proof of the complete algorithm.